- Batch – B5

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EEC23702 -Mini Project Work

EV-Agent: Agentic AI Powered Smart EV Charging Platform

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II -Project review on 22.09.26

Outline of the Presentation

- Introduction
- Objectives
- Literature Survey
- Existing System & Problems
- Scope of the Project
- Proposed Methodology
- Project Architecture
- Module & Architecture Component Explanation

Objectives

- **Optimize EV Charging** – Determine the best time and method to charge an electric vehicle efficiently.
- **Find Available Charging Stations** – Recommend suitable nearby charging stations based on availability and user requirements.
- **Reduce Charging Cost** – Suggest cost-effective charging schedules by considering electricity prices and demand.

Literature Survey

Author(s)	Year	Research Focus	Methodology	Drawback
Mishra et al.	**2021**	EV charging infrastructure in India	Reviewed development, infrastructure requirements and challenges of EV charging in India	No intelligent AI-based user recommendation
Ahmad et al.	**2022**	Optimal EV charging-station location	Reviewed station placement, charging demand, optimization and grid impact	Focuses mainly on infrastructure planning rather than real-time user assistance
Charging Technology & Recommendation Review	**2023**	Charging technologies & station recommendation	Reviewed charging technologies, infrastructure and station-finding/recommendation methods	Does not provide an Agentic AI system capable of reasoning and taking actions

Existing System & Problems

Existing System

- Manual Charging : Users manually select when and where to charge.
- Basic Charging Apps : Show charging stations and their availability.
- Fixed Scheduling : Charging is scheduled using predefined time slots.
- Limited AI : Existing systems provide basic recommendations.
- Separate Services : Cost, station selection, and charging are handled separately .

Problems

- High charging cost during peak hours.
- Charging stations may be unavailable when users arrive.
- No personalized charging plans.
- Limited real-time decision-making.
- Requires frequent user intervention.

Scope of the Project

- **Smart Charging** – Optimize charging time based on battery level and user needs.
- **Station Selection** – Identify suitable charging stations based on location and availability.
- **Cost Optimization** – Recommend cost-effective charging options.
- **Agentic AI** – Enable autonomous and intelligent charging decisions.
- **Personalized Planning** – Adapt charging plans according to user preferences.
- **Energy Efficiency** – Promote efficient energy usage and reduce peak demand.

Proposed Methodology

User & EV Details Input

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Battery Status Analysis

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Charging Station Data Collection

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User Requirement Analysis

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Charging Demand Prediction

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Station & Time Optimization

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Agentic AI Decision Making

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Cost & Energy Optimization

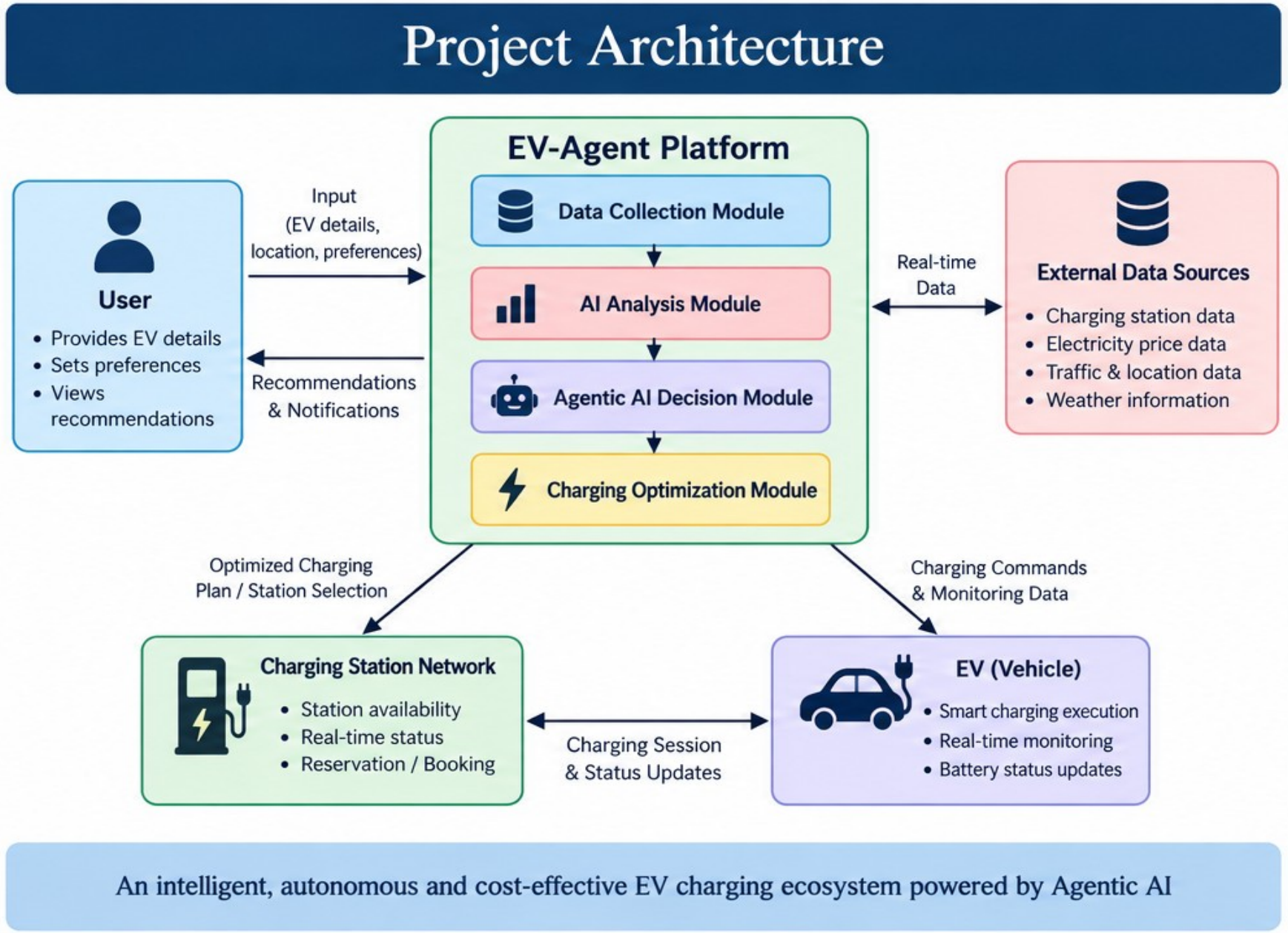
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Personalized Charging Recommendation

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Smart Charging & Real-Time Monitoring

Project Architecture



Module & Architecture Component Explanation

Module & Architecture Component Explanation

Module / Component	Description	Key Functions / Outputs
User	End user (EV owner) who provides vehicle details, location, and preferences through the web or mobile interface.	• Input EV details and travel needs • Set preferences (cost, time, station type) • Receive charging recommendations
Data Collection Module	Collects real-time and historical data from multiple sources.	• EV battery status • Charging station information • Electricity price data • Traffic, location and weather data
AI Analysis Module	Analyzes collected data using ML models to predict charging demand, station availability and optimal charging time.	• Predicts charging needs • Estimates station availability • Analyzes energy prices and demand
Agentic AI Decision Module	Uses Agentic AI to autonomously make decisions on when, where and how to charge based on user goals and real-time conditions.	• Plans optimal charging schedule • Selects best charging station • Adapts to real-time changes • Works with minimal user intervention

Continuation

Charging Optimization Module	Optimizes charging plan to minimize cost, charging time and energy consumption while meeting user requirements.	• Cost-effective charging schedule • Load balancing and peak time avoidance • Personalized recommendations
Charging Station Network	External network of EV charging stations that provide real-time availability and status information.	• Station availability and location • Real-time charging status • Reservation / booking support
EV (Vehicle)	Electric vehicle that receives charging commands and provides real-time battery status.	• Executes smart charging • Sends battery status and charging progress • Real-time monitoring and feedback
External Data Sources	Third-party data providers that supply real-time information to improve decision accuracy.	• Electricity price data • Traffic and location data • Weather information • Charging station APIs

Thank You!

For your attention

